AERODYNAMICS OF ROCKETS AND MISSILES (AAE 4081)

END SEMESTER ANSWER SCHEME – 29TH NOV 2024

Q1. What are the key differences between rockets and missiles with respect to the guidance systems and aerodynamic design?. (2)

Ans:

1. Guidance Systems

- **Rockets**:
  - Generally unguided or only partially guided, especially in the case of traditional rockets (e.g., fireworks, space launch vehicles).
  - In applications like space exploration, rockets often follow preprogrammed trajectories but do not adjust in real-time to dynamic targets.
  - Advanced rockets (e.g., space launchers) may include onboard inertial navigation and GPS systems for trajectory adjustments, but these are limited to following a planned route rather than seeking a target.
- **Missiles**:
  - Designed with sophisticated guidance systems that allow them to actively track and engage targets.
  - Guidance methods include:
    - **Inertial Navigation Systems (INS):** For basic course corrections.
    - **GPS or Satellite Guidance:** For precise navigation.
    - **Active/Passive Radar or Infrared Homing:** To track and engage moving targets in real time.
    - **Laser or Command Guidance:** Directed by an external source.
  - The ability to dynamically adjust its path makes a missile more versatile than a typical rocket in military or strategic applications.

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2. Aerodynamic Design

- **Rockets**:
  - Designed primarily for straight-line or ballistic trajectories.
  - Aerodynamic surfaces (if present) are typically optimized for stability during atmospheric flight but may be minimal or absent in rockets used in space (since there is no aerodynamic drag in space).
  - Focus on maximizing thrust and minimizing structural weight to achieve escape velocity or long-distance travel.
- **Missiles**:
  - Aerodynamically optimized for maneuverability within the atmosphere, allowing them to evade defenses or adjust their path toward moving targets.
  - Include control surfaces (e.g., fins, canards) and, in some cases, thrust-vectoring systems to enhance steering capabilities.
  - May sacrifice some range or payload capacity for the added complexity and weight of the guidance and control systems.

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Summary

- **Guidance:** Rockets are often unguided or follow preset trajectories, while missiles have advanced real-time guidance for tracking and hitting targets.

- **Aerodynamics:** Rockets prioritize stability and efficiency for their mission (often in a straight trajectory), while missiles prioritize maneuverability and precision.

These differences reflect their respective roles: rockets are primarily vehicles for transportation or bombardment, while missiles are designed as precision-targeting weapons.

Q2. Explain the missile control systems with the help of a neat diagram illustrating the six degrees of freedom, including yawning, rolling, and pitching movements of a missile. (3)

Ans:



The missile's control system operates within six degrees of freedom (DoF), which can be classified into three translational and three rotational motions. The equations governing these movements derive from Newton's Second Law for linear motion and Euler's Equations for rotational motion.

1. Translational Motion Equations

The missile's linear motion is described by the forces acting along its three principal axes (longitudinal, lateral, and vertical).

- Surge (Forward/Backward Motion):

$$F_x = m \cdot \frac{dU}{dt} + m \cdot W \cdot q - m \cdot V \cdot r$$

where:

- F_x: Force along the x-axis (thrust, drag).
- m: Mass of the missile.
- U, V, W: Velocity components along x, y, z axes.
- q, r: Angular velocities (pitch rate, yaw rate).

- **Sway (Side-to-Side Motion):**

$$F_y = m \cdot \frac{dV}{dt} + m \cdot U \cdot r - m \cdot W \cdot p$$

- **Heave (Up/Down Motion):**

$$F_z = m \cdot \frac{dW}{dt} + m \cdot V \cdot p - m \cdot U \cdot q$$

2. Rotational Motion Equations

The missile's rotational behavior is governed by **torques** acting about its three axes (roll, pitch, and yaw).

- **Roll (Rotation Along Longitudinal Axis):**

$$M_x = I_x \cdot \frac{dP}{dt} + (I_z - I_y) \cdot q \cdot r$$

where:

- M_x: Torque about the x-axis.
- P: Roll rate.
- I_x, I_y, I_z: Moments of inertia about the x, y, z axes.

- **Pitch (Rotation Along Lateral Axis):**

$$M_y = I_y \cdot \frac{dQ}{dt} + (I_x - I_z) \cdot r \cdot p$$

- **Yaw (Rotation Along Vertical Axis):**

$$M_z = I_z \cdot \frac{dR}{dt} + (I_y - I_x) \cdot p \cdot q$$

3. Coupled Dynamics

Since missiles operate in 3D space, these equations are interdependent. The control surfaces (e.g., fins, canards) generate forces and moments that influence both translational and rotational movements.

Q3. classify and explain the missiles based on the following criteria: (i) Launch Mode, (ii) Range, (iii) Warhead Type, (iv) Guidance Systems, and (v) Propulsion Systems.(5)

Ans:

Missiles can be classified based on various criteria, as described below:

(i) Launch Mode

Missiles are categorized based on the platform from which they are launched:

1. **Surface-to-Surface Missiles (SSM):** Launched from land or sea to target land-based or sea-based locations (e.g., Scud missile).
2. **Surface-to-Air Missiles (SAM):** Launched from land or sea to intercept aerial targets like aircraft or drones (e.g., Patriot missile).
3. **Air-to-Air Missiles (AAM):** Fired from aircraft to destroy other aircraft (e.g., AIM-120 AMRAAM).
4. **Air-to-Surface Missiles (ASM):** Launched from aircraft to attack ground targets (e.g., AGM-65 Maverick).
5. **Submarine-Launched Missiles (SLM):** Fired from submarines, targeting land or sea locations (e.g., Trident II).
6. **Anti-Tank Missiles:** Designed to destroy armored vehicles (e.g., Javelin).

(ii) Range ।ന

Missiles are classified by their operational range:

1. **Short-Range Missiles (SRBM):**
    - Range: Up to 1,000 km.
    - Example: Iskander missile.
2. **Medium-Range Missiles (MRBM):**
    - Range: Between 1,000 km and 3,500 km.
    - Example: Agni-II missile.
3. **Intermediate-Range Missiles (IRBM):**
    - Range: Between 3,500 km and 5,500 km.
    - Example: DF-26 missile.
4. **Intercontinental Ballistic Missiles (ICBM):**
    - Range: Over 5,500 km.
    - Example: Minuteman III.
5. **Tactical Missiles:**
    - Range: Generally less than 300 km.
    - Example: Hellfire missile.

(iii) Warhead Type ।ന

Missiles are categorized based on the type of payload they carry:

1. **Conventional Warheads:**
    - Contain explosives like TNT or RDX.
    - Example: Tomahawk cruise missile.
2. **Nuclear Warheads:**

- Designed to deliver a nuclear payload.
- Example: Trident II.

3. **Chemical or Biological Warheads:**
   - Deliver chemical or biological agents (prohibited under international law).
   - Rarely used.
4. **Cluster Warheads:**
   - Carry multiple smaller bombs that disperse over the target area.
   - Example: CBU-97 Sensor Fuzed Weapon.
5. **Penetration Warheads:**
   - Designed to penetrate hardened targets like bunkers.
   - Example: GBU-57 Massive Ordnance Penetrator.

(iv) Guidance Systems ١٣

Missiles are classified based on how they navigate to their target:

1. **Inertial Navigation System (INS):**
   - Uses onboard gyroscopes and accelerometers.
   - Example: V-2 rocket.
2. **Global Positioning System (GPS):**
   - Uses satellite signals for precise targeting.
   - Example: JDAM (Joint Direct Attack Munition).
3. **Infrared Homing:**
   - Tracks the heat signature of targets.
   - Example: AIM-9 Sidewinder.
4. **Radar Guidance:**
   - Can be **active** (missile carries its own radar) or **semi-active** (uses external radar).
   - Example: AMRAAM (active radar).
5. **Laser Guidance:**
   - Targets illuminated by a laser designator.
   - Example: Paveway bombs.
6. **Command Guidance:**
   - Controlled remotely from a ground station.
   - Example: Spike missile.

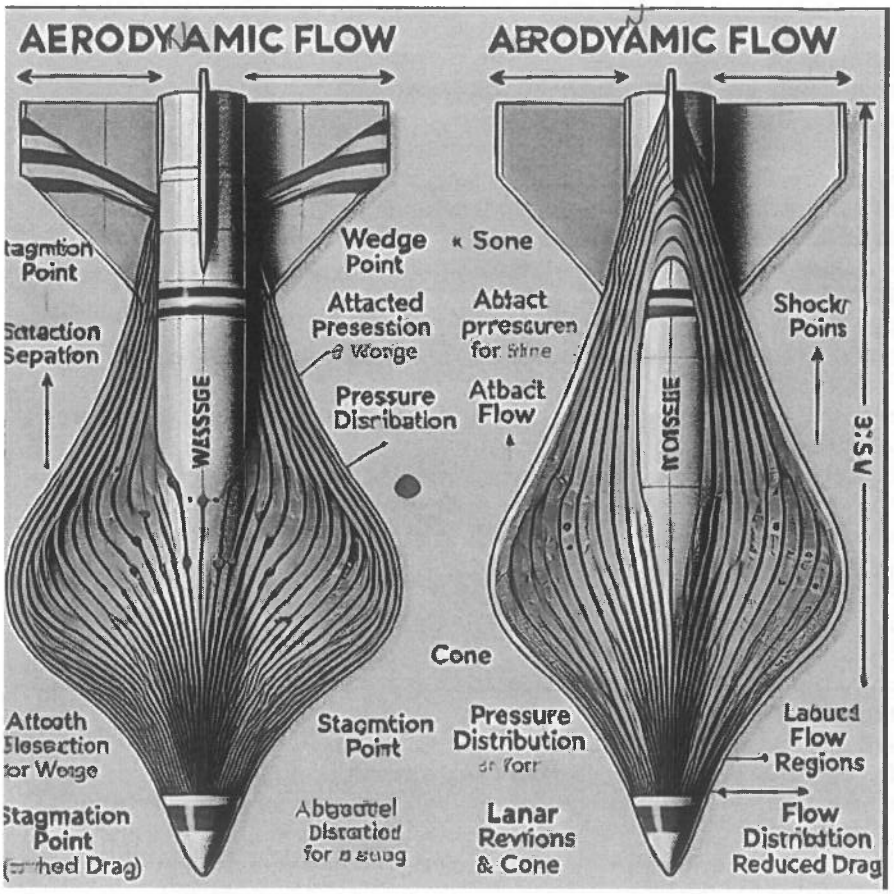
(v) Propulsion Systems ١٣

Missiles are classified by their propulsion technology:

1. **Solid-Propellant Missiles:**
    - Use solid fuel; simple, reliable, and quick to launch.
    - Example: Agni series.
2. **Liquid-Propellant Missiles:**
    - Use liquid fuel; provide higher energy but require more maintenance.
    - Example: Scud missile.
3. **Hybrid-Propellant Missiles:**
    - Combine features of both solid and liquid propellants.
    - Rarely used.
4. **Ramjet or Scramjet Missiles:**
    - Air-breathing engines for high-speed flight (ramjets up to Mach 5, scramjets beyond Mach 5).
    - Example: BrahMos (ramjet).
5. **Electric/Ion Propulsion:**
    - For long-duration missions in space.
    - Example: Not commonly used in military missiles but applicable to spacecraft.
6. **Turbojet or Turbofan Missiles:**
    - Use jet engines for long-range cruise capabilities.
    - Example: Tomahawk missile.

Q4. With the help of neat line diagrams, describe the aerodynamic flow characteristics over a missile nose for both wedge and cone shapes. (2)

Ans:



Here is a visual representation comparing the aerodynamic flow characteristics over a missile nose with wedge and cone shapes. The left side depicts the wedge shape with abrupt flow separations and high drag, while the right side shows the cone shape with smoother flow and reduced drag. Labels highlight key features such as stagnation points and shock waves.

Q5. Explain any three key differences between Homing Guidance, Beam Rider Guidance, and Inertial Navigation Guidance used in missile systems. (3)

Ans: Here are three key differences between Homing Guidance, Beam Rider Guidance, and Inertial Navigation Guidance used in missile systems:

1. Tracking Method

- **Homing Guidance:**
  - The missile actively tracks the target by detecting its emitted or reflected energy.
  - It uses sensors like infrared (IR), radar, or optical systems.
  - Example: Heat-seeking missiles like the AIM-9 Sidewinder rely on infrared homing.
- **Beam Rider Guidance:**
  - The missile follows a guidance beam emitted from a ground or airborne platform.

- The beam directs the missile to the target, with the missile staying within the beam's corridor.
- Example: Laser-guided missiles like the Starstreak.

- **Inertial Navigation Guidance:**
  - The missile uses onboard gyroscopes and accelerometers to calculate its position and trajectory relative to a preset path.
  - It does not require external signals or tracking once launched.
  - Example: ICBMs like the Minuteman III.

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2. Dependency on External Systems ١٣

- **Homing Guidance:**
  - Partially dependent on external systems; it requires the target to emit or reflect detectable signals.
  - For example, infrared homing requires the target to have a heat signature.
- **Beam Rider Guidance:**
  - Heavily dependent on external guidance beams.
  - If the beam is disrupted or the source is destroyed, the missile loses guidance.
- **Inertial Navigation Guidance:**
  - Fully independent of external systems after launch.
  - Immune to jamming or external interference but may accumulate errors over long distances.

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3. Suitability for Targets ١٣

- **Homing Guidance:**
  - Best for dynamic and moving targets (e.g., aircraft or ships).
  - The missile actively adjusts its trajectory based on the target's movements.
- **Beam Rider Guidance:**
  - Suitable for short-to-medium range engagements and direct line-of-sight targets.
  - Limited effectiveness against highly mobile or evasive targets.
- **Inertial Navigation Guidance:**
  - Best for long-range and stationary targets.
  - Not suitable for engaging moving targets, as it cannot adjust to changes in target position after launch.

Summary Table

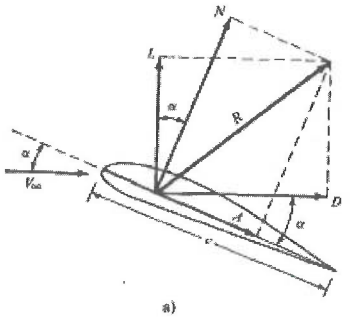
Feature	Homing Guidance	Beam Rider Guidance	Inertial Navigation Guidance
Tracking Method	Tracks target signals	Follows external beam	Relies on onboard sensors
External Dependency	Partially dependent	Highly dependent	Independent after launch
Target Suitability	Moving targets	Line-of-sight targets	Stationary or fixed targets

Each guidance method has specific advantages and limitations, making it suitable for different tactical scenarios.

Q6. Derive the expressions for aerodynamic forces and aerodynamic heating on a hypersonic missile vehicle, using the velocity-altitude map as a reference (5)

Ans:

Any object immersed in a flowing gas experiences a force as a result of the interaction of the body with the gas—the *aerodynamic force*, illustrated by the vector R in Fig. 1.21. The two components of R perpendicular and parallel to the relative wind far ahead of the body are *lift* L and *drag* D, respectively, as also shown in Fig. 1.21. Similarly, R can be resolved into two components perpendicular and parallel to the chord (or axis) of the body, the *normal force* N and the *axial force* A, respectively, shown in Fig. 1.21. These forces are usually



a)



$p = p(s)$ = surface pressure distribution
$\tau = \tau(s)$ = surface shear stress distribution

b)

Fig. 1.21 a) Resultant aerodynamic force and the components into which it splits and b) Illustration of pressure and shear stress on an aerodynamic surface.

couched in terms of force coefficients. Defining the dynamic pressure q_∞ as

$$q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2$$

where ρ_∞ and V_∞ are the freestream density and velocity respectively, and choosing some reference area S, we have the following by definition.

Lift coefficient:

$$C_L = \frac{L}{q_\infty S}$$

Drag coefficient:

$$C_D = \frac{D}{q_\infty S}$$

Normal-force coefficient:

$$C_N = \frac{N}{q_\infty S}$$

Axial-force coefficient:

$$C_A = \frac{A}{q_\infty S}$$

In addition, consider any point on the body. The resultant force R can create a moment M about this point, and we define

Moment coefficient:

$$\frac{M}{q_\infty S c}$$

where c is a characteristic length of the body, such as the chord length shown in Fig. 1.21a.

1.5 Hypersonic Flight Paths: Velocity-Altitude Map

Although this is a book on hypersonic and high-temperature gas dynamics, we must keep in mind that the frequent application of this material is to the design and understanding of hypersonic flight vehicles. In turn, it is helpful to have some knowledge of the flight paths of these vehicles through the atmosphere and the parameters that govern such flight paths. This is the purpose of the present section. In particular, we will examine the flight path of lifting and nonlifting hypersonic vehicles during atmospheric entry from space.

Consider a vehicle flying at a velocity V along a flight path inclined at the angle θ below the local horizontal, as shown in Fig. 1.22. The forces acting on the vehicle are lift L, drag D, and weight W; the thrust is assumed to be zero; hence, we are considering a hypersonic glide vehicle. Summing forces along and perpendicular to the curvilinear flight path, we obtain the following equations of motion from Newton's second law.

Along flight path:

$$W \sin \theta - D = m \frac{dV}{dt} \tag{1.3}$$

Perpendicular to flight path:

$$L - W \cos \theta = -m \frac{V^2}{R} \tag{1.4}$$

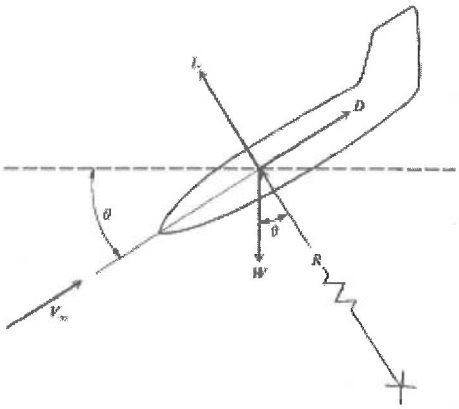


Fig. 1.22 Force diagram for reentry body.

In Eq. (1.4), R is the local radius of curvature of the flight path. For most entry conditions, θ is small; hence, we assume $\sin\theta \approx 0$ and $\cos\theta \approx 1$. For this case, Eqs. (1.3) and (1.4) become, noting that $m = W/g$

$$-D = \frac{W}{g}\frac{dV}{dt} \tag{1.5}$$

$$L - W = -\frac{W}{g}\frac{V^2}{R} \tag{1.6}$$

The drag can be expressed in terms of the drag coefficient C_D as $D = \frac{1}{2}\rho V^2 S C_D$, where ρ is the freestream density and S is a reference area. Hence, Eq. (1.5) becomes

$$-\frac{1}{2}\rho V^2 S C_D = \frac{W}{g}\frac{dV}{dt}$$

Rearranging, we obtain

$$-\frac{1}{g}\frac{dV}{dt} = \left(\frac{W}{C_D S}\right)^{-1}\frac{\rho V^2}{2} \tag{1.7}$$

In Eq. (1.7), $W/C_D S$ is defined as the ballistic parameter; it clearly influences the flight path of the entry vehicle via the solution of Eq. (1.7). For a purely ballistic reentry (no lift), $W/C_D S$ is the only parameter governing the flight path for a given entry angle.

Returning to Eq. (1.6) and expressing the lift in terms of the lift coefficient C_L as $L = \frac{1}{2}\rho V^2 S C_L$, we obtain

$$\frac{1}{2}\rho V^2 S C_L - W = -\frac{W}{g}\frac{V^2}{R}$$

Rearranging,

$$1 - \frac{1}{g}\frac{V^2}{R} = \left(\frac{W}{C_L S}\right)^{-1}\frac{\rho V^2}{2} \tag{1.8}$$

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Q7. Discuss the working principles and features of the following missile warheads: (i) Shaped Charge Warhead, (ii) Fragmentation Warhead. (2)

Ans:

(i) Shaped Charge Warhead 1m

Working Principle:

- A **shaped charge warhead** is designed to focus explosive energy into a narrow, high-velocity jet to penetrate heavily armored targets.
- The warhead typically consists of a **conical or hemispherical liner** (made of metal like copper) placed in front of an explosive charge.
- When the explosive detonates, the liner collapses and forms a **metallic jet** that moves at extremely high velocity (up to 10 km/s).

- This jet creates intense pressure and heat, allowing it to pierce through thick armor or reinforced structures.

Features:

1. **Armor Penetration**:
    - Highly effective against tanks, armored vehicles, and bunkers.
    - Can penetrate armor several times thicker than the warhead's diameter (e.g., 5-10 times).
2. **Directional Energy**:
    - Focuses nearly all explosive energy in one direction, reducing collateral damage.
3. **Insensitive to Angle of Impact**:
    - Works effectively even when striking at an angle, as the jet maintains its penetration capability.
4. **Applications**:
    - Used in anti-tank guided missiles (e.g., TOW missile), RPGs, and some air-to-surface missiles.

(ii) Fragmentation Warhead

Working Principle:

- A **fragmentation warhead** disperses high-speed fragments (shrapnel) upon detonation to maximize damage over a wide area.
- The warhead casing is designed to fragment upon explosion, or it may contain pre-formed fragments (e.g., steel balls, rods).

- When the explosive charge detonates, these fragments are propelled outward at high velocities (typically 1.5-2 km/s).
- The damage is caused by the kinetic energy of the fragments striking the target.

Features:

1. **Area Damage**:
    - Effective against soft targets such as infantry, vehicles, and unarmored structures.
    - Covers a larger area compared to shaped charges.
2. **Pre-Formed Fragments**:
    - Modern designs use precisely engineered fragments for consistent size, shape, and dispersion patterns.
3. **Proximity Effect**:
    - Often paired with proximity fuzes to detonate near the target for optimal fragment dispersion.
4. **Applications**:
    - Used in surface-to-air missiles, anti-aircraft missiles (e.g., Patriot missile), and air-to-ground bombs for wide-area damage.

Comparison

Feature	Shaped Charge Warhead	Fragmentation Warhead
Damage Mechanism	High-velocity jet	High-speed fragments
Primary Target	Armored vehicles, bunkers	Soft targets, infantry, aircraft
Blast Direction	Focused	Omnidirectional or controlled area

Feature	Shaped Charge Warhead	Fragmentation Warhead
Collateral Damage	Low	Moderate to high
Key Application	Anti-tank, anti-armor	Anti-personnel, anti-aircraft

Each type of warhead is tailored for specific mission requirements, balancing effectiveness, precision, and area coverage.

Q8. Elucidate the operation and uses of the following missile fuzes: (i) Impact Fuze, (ii) Altitude Fuze, (iii) Proximity Fuze. (3)

(i) Impact Fuze ١م

Operation:

- An **impact fuze** is triggered when the missile makes physical contact with the target.
- It typically consists of a mechanical, electrical, or chemical device that activates the warhead upon impact.
- The fuze can be designed for:
  - **Direct Impact:** Detonates immediately upon contact with the target surface.
  - **Delay Impact:** Delays detonation slightly to allow the missile to penetrate the target (e.g., hardened structures).

Uses:

- Ideal for **direct strikes** on targets such as:
  - Tanks and armored vehicles.
  - Buildings or fortifications.
- Frequently used in **anti-armor missiles** or bombs where penetration is required before explosion.
- Example: Anti-tank guided missiles like the **Javelin** often use impact fuzes.

(ii) Altitude Fuze ١م

Operation:

- An **altitude fuze** detonates the missile warhead at a predetermined height above the ground or target.
- It uses barometric sensors, radar altimeters, or laser rangefinders to measure altitude.
- Detonation occurs when the missile reaches the set altitude, regardless of whether it has struck the target.

Uses:

- Effective for creating wide-area effects, such as:
  - **Airburst Explosions:** Maximizing damage to infantry and unarmored vehicles over a larger area.
  - **Cluster Bombs:** Dispersing submunitions at a precise height.
- Commonly used in **surface-to-surface missiles** or bombs where ground targets need to be neutralized with high efficiency.
- Example: Artillery shells or cluster munitions using altitude fuzes.

(iii) Proximity Fuze ١م

Operation:

- A **proximity fuze** detonates the warhead when the missile is near the target rather than upon direct contact.

- It uses sensors such as radar, infrared, or acoustic systems to detect the target's presence and calculate the optimal detonation moment.
- The fuze creates a spherical or directional blast to ensure maximum damage to the target.

Uses:

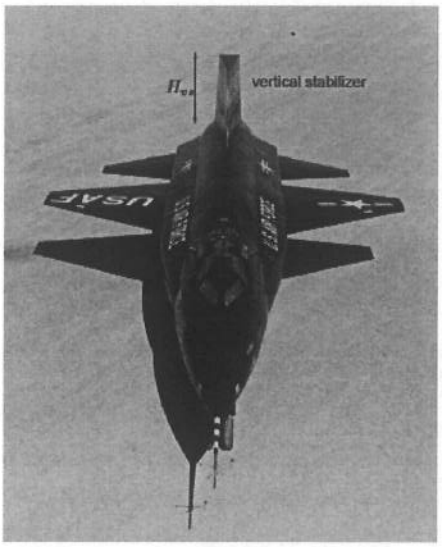
- Highly effective for engaging **moving or airborne targets** like:
  - Aircraft and drones.
  - Missiles or projectiles in mid-flight.
- Also used in **anti-personnel and anti-vehicle roles**, ensuring a blast at an optimal range.
- Example: The **Patriot missile** uses proximity fuzes for intercepting and destroying incoming threats.

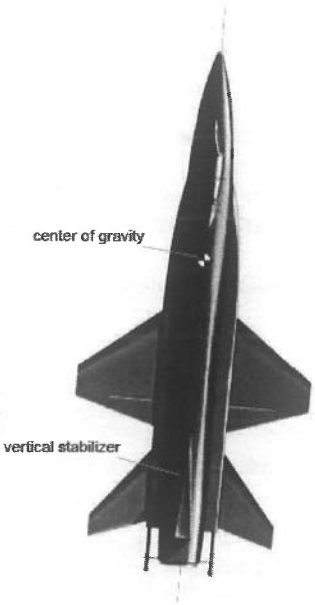
Comparison Table

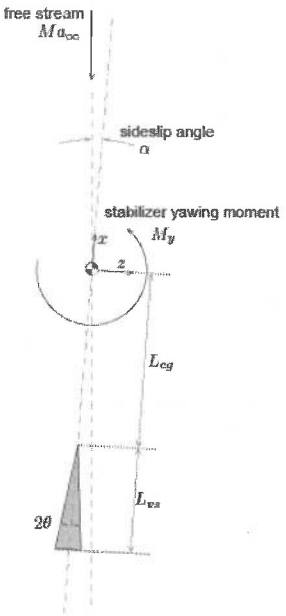
Feature	**Impact Fuze**	**Altitude Fuze**	**Proximity Fuze**
Trigger Mechanism	Physical contact	Pre-set altitude	Target detection at close range
Detonation Timing	On impact or slightly delayed	At a predetermined height	Near the target
Primary Targets	Armored vehicles, bunkers	Infantry, soft targets, area damage	Aircraft, drones, incoming missiles
Common Applications	Anti-tank, bunker-busting missiles	Airburst shells, cluster bombs	Surface-to-air and anti-aircraft missiles

Each fuze type is designed for specific tactical needs, ensuring maximum effectiveness in various combat scenarios.

Q9. A vertical stabilizer is a rudder mounted on the aft end of the fuselage that provides directional stability to the aircraft when it sideslips with respect to the free stream. For instance, the X-15 hypersonic rocket plane used a vertical stabilizer that had the shape of a wedge of semi-angle θ. When the rocket plane side slipped by an angle α, the aerodynamic forces on the vertical stabilizer induced a yawing moment M_y around the center of gravity that restored the lateral attitude. In the notation below, L_{cg} is the distance from the vertical stabilizer's edge to the center of gravity, whereas L_{vs} and H_{vs} are, respectively, the length and height of the vertical stabilizer.







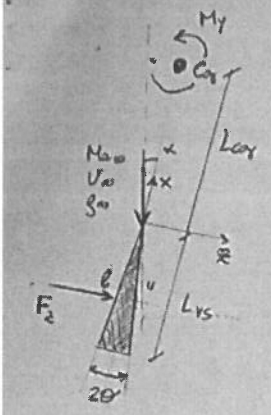
$$C_y = \frac{2M_y}{\rho_\infty U_\infty^2 H_{vs} L_{vs} L_{cg}}$$

Assuming that the free-stream Mach number is $Ma_\infty = 8$, and the sideslip angle is $\alpha = 12^\circ$, calculate the yawing moment coefficient in the following cases:

a) $\theta = 20^\circ$.
b) $\theta = 8^\circ$.
c) Redo parts (a) and (b) using the Newtonian theory of hypersonic and compare the solutions. (5)

Ans:

SOLUTION



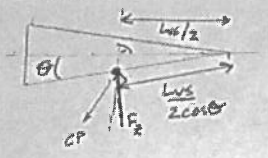
* SIDEWAYS (LIFT) FORCE ON THE VS:

$$F_2 = \frac{1}{2}\rho_\infty U_\infty^2 \left(C_{P\ell} - C_{Pu}\right) \frac{L_{VS}}{\cos\theta} H_{VS} \cos\theta$$

WITH $C_{P\ell} = \dfrac{P_\ell - P_\infty}{\frac{1}{2}\rho_\infty U_\infty^2}$ AND $C_{Pu} = \dfrac{P_u - P_\infty}{\frac{1}{2}\rho_\infty U_\infty^2}$

THE PRESSURE COEFFICIENTS ON THE LOWER AND UPPER SURFACE

* YAWING MOMENT: $M_y = F_2 \left(L_{cg} + \underbrace{\frac{L_{VS}}{2}}_{\text{LINE PASSING THROUGH THE CENTER OF PRESSURE}}\right) \overset{L_{VS}/L_{cg} \ll 1}{\approx} F_2 L_{cg}$



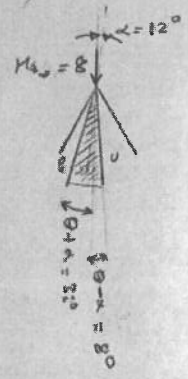
* YAWING MOMENT COEFFICIENT

$$C_y = \frac{M_y}{\frac{\rho_\infty U_\infty^2}{2} L_{VS} L_{cg} H_{VS}} = \frac{\frac{1}{2}\rho_\infty U_\infty^2 \left(C_{P\ell} - C_{Pu}\right) L_{VS} H_{VS} L_{cg}}{\frac{1}{2}\rho_\infty U_\infty^2 L_{VS} L_{cg} H_{VS}}$$

$$= \left(C_{P\ell} - C_{Pu}\right) \quad (1)$$

1m

a) $\theta = 20^\circ$



$Ma_\infty = 8$, $\delta_\ell = \theta + \alpha = 32^\circ$ } OBLIQUE SHOCK CHART: $\beta_\ell = 42^\circ$

$$\Rightarrow \frac{P_\ell}{P_\infty} = 1 + \frac{2\gamma}{\gamma+1}\left(Ma_\infty^2 \sin^2(\beta_\ell - \Delta)\right) = 33.2$$

$$C_{P\ell} = \frac{2}{\gamma Ma_\infty^2}\left(\frac{P_\ell}{P_\infty} - 1\right) = \underline{0.72}$$

$Ma_\infty = 8$, $\delta_u = \theta - \alpha = 8^\circ$ } OBLIQUE SHOCK CHART $\beta_u = 12^\circ$

$$\Rightarrow \frac{P_u}{P_\infty} = 1 + \frac{2\gamma}{\gamma+1}\left(Ma_\infty^2 \sin^2\beta_u - 1\right) = 3.1$$

$$C_{Pu} = \frac{2}{\gamma Ma_\infty^2}\left(\frac{P_u}{P_\infty} - 1\right) = \underline{0.05}$$

$\Rightarrow$ SUBSTITUTING $C_{P\ell}$ AND C_{Pu} INTO (1): $C_y = \underline{0.67}$

2m

FROM NEWTONIAN THEORY:

$$C_{P\ell} = 2\sin^2\delta_\ell = 0.58$$
$$C_{Pu} = 2\sin^2\delta_u = 0.04$$
$$\Rightarrow C_y = \underline{0.52}$$

b) $\theta = 8^\circ$



$Ma_\infty = 8$

$\delta_2 = \theta + \alpha = 20^\circ$ } OBLIQUE CHART : $\beta_2 = 26.5^\circ$

$$\frac{P_2}{P_\infty} = 1 + \frac{2\gamma}{\gamma+1}\left(Ma_\infty^2 \sin^2\beta_2 - 1\right) = 14.8$$

$$C_{P_2} = \frac{2}{\gamma Ma_\infty^2}\left(\frac{P_2}{P_\infty} - 1\right) = 0.31$$

$\delta_U = \alpha - \theta = 4^\circ$ }
$Ma_\infty = 8$ } FROM TABLES OF PRANDTL-MEYER EXPANSION FAN

$\nu(Ma_\infty) = 95.6^\circ$

$\Rightarrow \nu(Ma_U) = \nu(Ma_\infty) + \delta_U = 99.6^\circ \Rightarrow Ma_U \simeq 9.1$

THEREFORE $$\frac{P_U}{P_\infty} = \left(\frac{1 + \left(\frac{\gamma-1}{2}\right)Ma_\infty^2}{1 + \left(\frac{\gamma-1}{2}\right)Ma_U^2}\right)^{\frac{\gamma}{\gamma-1}} = 0.43$$

$$\Rightarrow C_{P_U} = \frac{2}{\gamma Ma_\infty^2}\left(\frac{P_U}{P_\infty} - 1\right) = -4.8 \cdot 10^{-3}$$

SUBSTITUTING C_{P_U} AND C_{P_2} INTO (1) : $C_y = 0.31$

FROM THE NEWTONIAN THEORY : $C_{P_2} = 2\sin^2\delta_2 = 0.23$ } $C_y \simeq 0.23$
$C_{P_U} = 0$

2m

Q10. If the vertical component of the landing velocity of a parachute is equal to that acquired during a free fall of 2 m, find the diameter of a open parachute (hollow and hemisphere) if the total weight of the parachute and the person is 950 N. Assume at the ambient conditions, density = 1.2 kg/m^3 and C_d=1.35. (2)

Ans:



$V_2^2 = V_1^2 + 2as$

$V = \sqrt{2x9.81x2} = 6.264\ m/s$

$NSL:$

$\sum F = 0$

$m.g - \frac{1}{2}\rho.V^2.A_f x C_d = 0$ } 1m

$950 - \frac{1}{2}x1.2x6.264^2.(\frac{\pi}{4})D^2x1.4 = 0$

$solve \rightarrow D = 6.17\ m$ } 1m

Q11. Airflow over a sharp edge missile flat plate, 3 m long and 3 m wide at a velocity of 2 m/s. Determine Drag force if (i) the plate is kept horizontal and (ii) the plate was mounted perpendicular to the flow direction. Assume C_d=1.4 for both the cases. For air, take density as 1.23 kg/m^3, and kinematic viscosity 1.46x10^{-05} m/s^2. [3]

Solution:

1.

$$Re_l = \frac{V_\infty xL}{\upsilon} = \frac{2x3}{1.46x10^{-5}} = 4.11x10^5 < 5x10^5 \rightarrow Laminar$$

$$C_D = \frac{1.328}{(Re)^{0.5}} = 0.00207$$

$$F_D = \frac{1}{2}\rho.A.V_\infty^2 x C_D = \frac{1}{2}x1.23x(3x3)x2^2x0.00207 = 0.045\ N$$ } 1½m

2.

$Cd = 1.4$

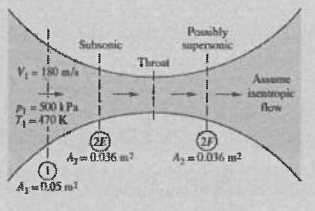
$$F_D = \frac{1}{2}\rho.A.V_\infty^2 x C_D = \frac{1}{2}x1.23x(3x3)x2^2x1.4 = 30.4\ N$$ } 1½m

Q12. Air flows isentropically through a missile duct. At section 1 the area is 0.05 m^2 and V_1 = 180 m/s, p_1 = 500 kPa, and T_1 = 470 K. Compute (*a*) T_0, (*b*) M_{a1}, (*c*) p_0, and (*d*) both A^* and mass flow rate. If at section 2 the area is 0.036 m^2, compute M_{a2} and p_2 if the flow is (*e*) subsonic or (f) supersonic. Assume Y = 1.4. [5]

Ans:

Solution

A general sketch of the problem is shown in Fig. E9.4. With V_1 and T_1 known, the energy equation (9.23) gives



E9.4

$$T_0 = T_1 + \frac{V_1^2}{2c_p} = 470 + \frac{(180)^2}{2(1005)} = 486\ \text{K} \qquad \textit{Ans. (a)}$$

The local sound speed $a_1 = \sqrt{kRT_1} = [(1.4)(287)(470)]^{1/2} = 435$ m/s. Hence

$$\text{Ma}_1 = \frac{V_1}{a_1} = \frac{180}{435} = 0.414 \qquad \textit{Ans. (b)}$$

With Ma_1 known, the stagnation pressure follows from Eq. (9.34):

$$p_0 = p_1(1 + 0.2\,\text{Ma}_1^2)^{3.5} = (500\ \text{kPa})[1 + 0.2(0.414)^2]^{3.5} = 563\ \text{kPa} \qquad \textit{Ans. (c)}$$

Similarly, from Eq. (9.45), the critical sonic throat area is

$$\frac{A_1}{A^*} = \frac{(1 + 0.2\,\text{Ma}_1^2)^3}{1.728\,\text{Ma}_1} = \frac{[1 + 0.2(0.414)^2]^3}{1.728(0.414)} = 1.547$$

or

$$A^* = \frac{A_1}{1.547} = \frac{0.05\ \text{m}^2}{1.547} = 0.0323\ \text{m}^2 \qquad \textit{Ans. (d)}$$

This throat must *actually be present* in the duct if the flow is to become supersonic.

We now know A^*. So to compute the mass flow we can use Eqs. (9.46), which remain valid, based on the numerical value of A^*, whether or not a throat actually exists:

$$\dot{m} = 0.6847\frac{p_0 A^*}{\sqrt{RT_0}} = 0.6847\frac{(563{,}000)(0.0323)}{\sqrt{(287)(486)}} = 33.4\ \text{kg/s} \qquad \textit{Ans. (d)}$$

Or we could fare equally well with our new "local mass flow" formula, Eq. (9.47), using, say, the pressure and area at section 1. Given $p_1/p_0 = 500/563 = 0.889$, Eq. (9.47) yields

$$\frac{\dot{m}\sqrt{287(486)}}{563{,}000(0.05)} = \sqrt{\frac{2(1.4)}{0.4}}(0.889)^{2/1.4}[1 - (0.889)^{0.4/1.4}] = 0.444 \quad \dot{m} = 33.4\frac{\text{kg}}{\text{s}} \qquad \textit{Ans. (d)}$$

Assume *subsonic* flow corresponds to section 2E in Fig. E9.4. The duct contracts to an area ratio $A_2/A^* = 0.036/0.0323 = 1.115$, which we find on the left side of Fig. 9.7 or the subsonic part of Table B.1. Neither the figure nor the table is that accurate. EES will give an arbitrarily accurate solution with only three statements (in SI units):

```
A2 = 0.036
Astar = 0.0323
A2/Astar = (1 + 0.2*Ma2^2)^3/1.2^3/Ma2
```

Specify that you want a *subsonic* solution (e.g., limit $\text{Ma}_2 < 1$), and EES reports

$$\text{Ma}_2 = 0.6758 \qquad \textit{Ans. (e)}$$

[Ask for a supersonic solution (require $\text{Ma}_2 > 1$) and you receive $\text{Ma}_2 = 1.4001$, which is the answer to part (*f*).] The pressure is given by the isentropic relation

$$p_2 = \frac{p_0}{[1 + 0.2(0.676)^2]^{3.5}} = \frac{563\ \text{kPa}}{1.358} = 415\ \text{kPa} \qquad \textit{Ans. (e)}$$

Part (*e*) does *not* require a throat, sonic or otherwise; the flow could simply be contracting subsonically from A_1 to A_2.

This time assume *supersonic* flow, corresponding to section 2F in Fig. E9.4. Again the area ratio is $A_2/A^* = 0.036/0.0323 = 1.115$, and we look on the *right* side of Fig. 9.7 or the supersonic part of Table B.1—the latter can be read quite accurately as $\text{Ma}_2 = 1.40$. EES will give a very accurate solution with the same three statements from part (*e*). Specify that you want a *supersonic* solution (e.g., limit $\text{Ma}_2 > 1$), and EES reports

$$\text{Ma}_2 = 1.4001 \qquad \textit{Ans. (f)}$$

Again the pressure is given by the isentropic relation at the new Mach number.

$$p_2 = \frac{p_0}{[1 + 0.2(1.4001)^2]^{3.5}} = \frac{563\ \text{kPa}}{3.183} = 177\ \text{kPa} \qquad \textit{Ans. (f)}$$

Note that the supersonic flow pressure level is much less than p_2 in part (*e*), and a sonic throat *must* have occurred between sections 1 and 2F.

Q13. A low flying missile develops a nose temperature of 2500 K when the ambient temperature and pressure are 250 K and 0.01 bar respectively. Determine the missile velocity and its stagnation pressure. Assume for air $\Upsilon = 1.4$ and $C_p = 1005$ J/kg K. (2)

A low flying missile develops a nose temperature of 2500K when the ambient temperature and pressure are 250K and 0.01 bar respectively. Determine the missile velocity and its stagnation pressure. Assume for air: $\gamma = 1.4$. Cp = 1005 J/kgK

Ans: [2126 m/s, 31.6 bar]

Solution:

$$Cp = Cv + R$$

$$Cp = Cp/\gamma + R$$

hence

$$R = \frac{(\gamma - 1)}{\gamma} xCp = \frac{1.4 - 1}{1.4} x1005 = 287.143\ J/kgK$$

$$T_o - T = V^2/2Cp.$$

$$V = \sqrt{2xCpx(T_o - T)} = \sqrt{2x1005x(2500 - 250)} = 2126.67\ m/s$$

$$\frac{P_o}{P} = [1 + \frac{V^2}{2.Cp.T}]^{\frac{\gamma}{\gamma-1}}$$

$$\frac{P_o}{P} = [1 + \frac{V^2}{2.Cp.T}]^{\frac{\gamma}{\gamma-1}}$$

$$P_o = Px\left[1 + \frac{V^2}{2Cp.T}\right]^{\frac{\gamma}{\gamma-1}}$$

$$= 0.01x\left[1 + \frac{2126.67^2}{2x1005x(273 + 250)}\right]^{\frac{1.4}{0.4}} = 31.6\ bar$$

Q14. An airplane wing has a 7.62 m span and 2.13 m chord. Estimate the drag on the wing (two sides) treating it as a flat plate and flight speed of 89.4 m/s to be turbulent from the leading edge onward.

Determine the reduction in power that can be saved if the boundary control device is installed on the wing to ensure laminar flow over to the entire wing surface. For air take density as 1.01 kg/m^3 and kinematic viscosity as 1.3x10^{-05} m/s^2. (3)

Ans:

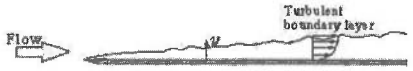
3.6 (a) An airplane wing has a 7.62m span and 2.13m chord. Estimate the drag on the wing (two sides) treating it as a flat plate and the flight speed of 89.4 m/s to be turbulent from the leading edge onward.

(b) Determine the reduction in power that can be saved if the boundary layer control device is installed on the wing to ensure laminar flow over the entire wing's surface.

For air, take density as 1.01 kg/m³, and kinematic viscosity as 1.3×10^{-5} m/s².

Ans (358 N, 86%)

Solution:



$$C_D = \frac{1.328}{\sqrt{Re_L}} \quad \text{laminar,} \quad Re_L < 5\times10^5$$

$$C_D = \frac{0.074}{[Re_L]^{\frac{1}{5}}} \quad \text{turbulent,} \quad 5\times10^5 \le Re_L < 10^7$$

a)

$$Re_l = \frac{V_\infty xL}{\upsilon} = \frac{89.4x2.13}{1.3x10^{-5}} = 1.265x10^7 > 5x10^5 \rightarrow Turbulent$$

$$C_D = \frac{0.074}{(Re)^{0.2}} = 0.00273$$

2m

$$F_D = \frac{1}{2}\rho.A.V_\infty{}^2 xC_D = \frac{1}{2}x1.01x(2x7.62x2.13)x89.4^2 x0.00273 = 357.6\ N$$

$$Power = F_D xV = 357.6x89.4 = 31.969\ kW$$

b) *laminar.*

$$C_D = \frac{1.328}{(1.265x10^7)^{0.5}} = 0.00037$$

$$F_D = \frac{1}{2}\rho.A.V_\infty{}^2 xC_D = \frac{1}{2}x1.01x(2x7.62x2.13)x89.4^2 x0.00037 = 48.9\ N$$

1m

$$Power = F_D xV = 48.9x89.4 = 4.372\ kW$$

$$\%reduction = 86.3\%$$

Q15. The velocity profile of a turbulent boundary layer flow is given as $\frac{u}{U} = \left(\frac{y}{\delta}\right)^7$. Find an expression for boundary layer thickness, shear stress, and drag force on one side of the plate and coefficient of drag in terms of Reynolds number. Given the shear stress (τ_0) for turbulent boundary layer as:

$$\tau_0 = 0.0225\,\rho\,U^2\left(\frac{\mu}{\rho U\delta}\right)^{\frac{1}{4}} \quad (5)$$

Sol. Given : $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7}$

Substituting this value in Von Kármán momentum integral equation (13.10),

$$\frac{\tau_0}{\rho U^2} = \frac{\partial}{\partial x}\left[\int_0^\delta \frac{u}{U}\left(1 - \frac{u}{U}\right) dy\right]$$

$$= \frac{\partial}{\partial x}\left[\int_0^\delta \left(\frac{y}{\delta}\right)^{1/7}\left[1 - \left(\frac{y}{\delta}\right)^{1/7}\right] dy\right] = \frac{\partial}{\partial x}\left[\int_0^\delta \left(\frac{y^{1/7}}{\delta^{1/7}} - \frac{y^{2/7}}{\delta^{2/7}}\right) dy\right]$$

$$= \frac{\partial}{\partial x}\left[\frac{y^{\frac{1}{7}+1}}{\left(\frac{1}{7}+1\right)\delta^{1/7}} - \frac{y^{\frac{2}{7}+1}}{\left(\frac{2}{7}+1\right)\delta^{2/7}}\right]_0^\delta$$

$$= \frac{\partial}{\partial x}\left[\frac{7}{8}\frac{y^{8/7}}{\delta^{1/7}} - \frac{7}{9}\frac{y^{9/7}}{\delta^{2/7}}\right]_0^\delta = \frac{\partial}{\partial x}\left[\frac{7}{8}\frac{\delta^{8/7}}{\delta^{1/7}} - \frac{7}{9}\frac{\delta^{9/7}}{\delta^{2/7}}\right]$$

$$= \frac{\partial}{\partial x}\left[\frac{7}{8}\delta - \frac{7}{9}\delta\right] = \frac{\partial}{\partial x}\left[\frac{63-56}{72}\right]\delta = \frac{\partial}{\partial x}\left[\frac{7}{72}\right]\delta = \frac{7}{72}\frac{\partial \delta}{\partial x}$$

In the above expression, the integration limits should be from δ' to δ. But as the laminar sub-layer is very thin that is δ' is very small. Hence the limits of integration are taken from 0 to δ.

Now $$\tau_0 = \frac{7}{72}\rho U^2 \frac{\partial \delta}{\partial x} \quad ...(13.38)$$

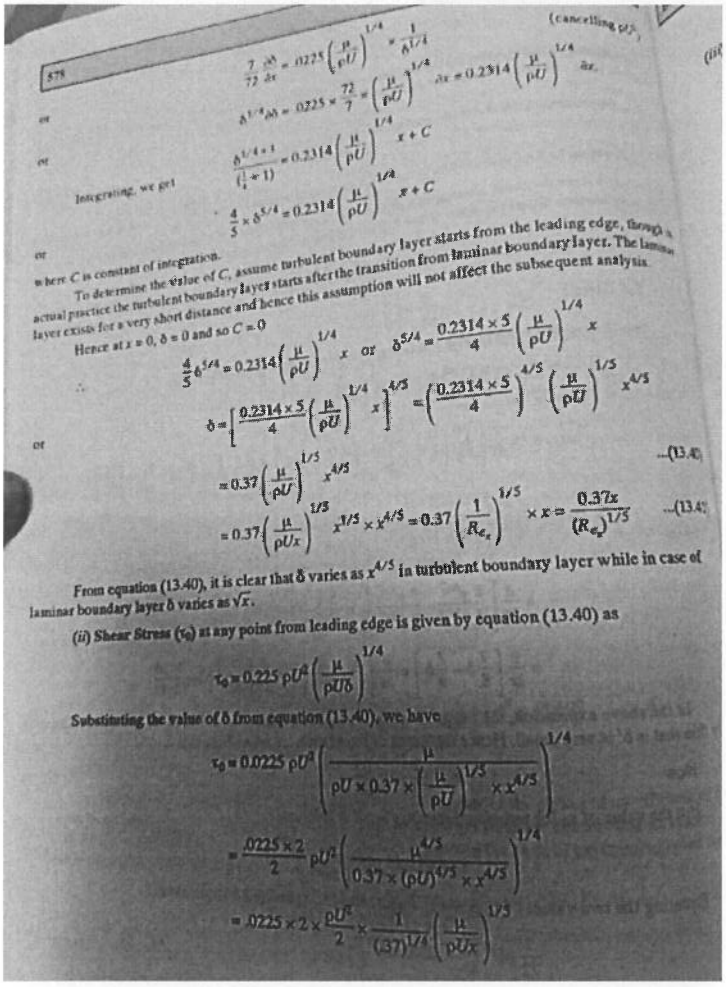
But the value of τ_0 for turbulent boundary layer is given,

$$\tau_0 = 0.0225\,\rho U^2\left(\frac{\mu}{\rho U \delta}\right)^{1/4} \quad ...(13.39)$$

Equating the two values of τ_0 given by equations (13.38) and (13.39), we have

$$\frac{7}{72}\rho U^2 \frac{\partial \delta}{\partial x} = .0225\,\rho U^2\left(\frac{\mu}{\rho U \delta}\right)^{1/4}$$

1m



578

$$\frac{7}{72}\frac{\partial \delta}{\partial x} = .0225\left(\frac{\mu}{\rho U}\right)^{1/4}\frac{1}{\delta^{1/4}} \quad \text{(cancelling } \rho U^2\text{)}$$

or $$\delta^{1/4}\partial\delta = .0225 \times \frac{72}{7}\left(\frac{\mu}{\rho U}\right)^{1/4} \partial x = 0.2314\left(\frac{\mu}{\rho U}\right)^{1/4} \partial x.$$

or $$\frac{\delta^{1/4+1}}{\left(\frac{1}{4}+1\right)} = 0.2314\left(\frac{\mu}{\rho U}\right)^{1/4} x + C$$

Integrating, we get $$\frac{4}{5}\times\delta^{5/4} = 0.2314\left(\frac{\mu}{\rho U}\right)^{1/4} x + C$$

or

where C is constant of integration.

To determine the value of C, assume turbulent boundary layer starts from the leading edge, though in actual practice the turbulent boundary layer starts after the transition from laminar boundary layer. The laminar layer exists for a very short distance and hence this assumption will not affect the subsequent analysis.

Hence at $x = 0$, $\delta = 0$ and so $C = 0$

$$\frac{4}{5}\delta^{5/4} = 0.2314\left(\frac{\mu}{\rho U}\right)^{1/4} x \quad \text{or} \quad \delta^{5/4} = \frac{0.2314 \times 5}{4}\left(\frac{\mu}{\rho U}\right)^{1/4} x$$

$$\therefore \quad \delta = \left\{\frac{0.2314 \times 5}{4}\left(\frac{\mu}{\rho U}\right)^{1/4} x\right\}^{4/5} = \left(\frac{0.2314 \times 5}{4}\right)^{4/5}\left(\frac{\mu}{\rho U}\right)^{1/5} x^{4/5}$$

or

$$= 0.37\left(\frac{\mu}{\rho U}\right)^{1/5} x^{4/5} \quad ...(13.40)$$

$$= 0.37\left(\frac{\mu}{\rho U x}\right)^{1/5} x^{1/5} \times x^{4/5} = 0.37\left(\frac{1}{R_{e_x}}\right)^{1/5} \times x = \frac{0.37x}{(R_{e_x})^{1/5}} \quad ...(13.41)$$

2m

From equation (13.40), it is clear that δ varies as $x^{4/5}$ in turbulent boundary layer while in case of laminar boundary layer δ varies as $\sqrt{x}$.

(ii) **Shear Stress (τ_0)** at any point from leading edge is given by equation (13.40) as

$$\tau_0 = 0.225\,\rho U^2\left(\frac{\mu}{\rho U \delta}\right)^{1/4}$$

Substituting the value of δ from equation (13.40), we have

$$\tau_0 = 0.0225\,\rho U^2\left(\frac{\mu}{\rho U \times 0.37 \times \left(\frac{\mu}{\rho U}\right)^{1/5} \times x^{4/5}}\right)^{1/4}$$

$$= \frac{.0225 \times 2}{2}\rho U^2\left(\frac{\mu^{4/5}}{0.37 \times (\rho U)^{4/5} \times x^{4/5}}\right)^{1/4}$$

$$= .0225 \times 2 \times \frac{\rho U^2}{2} \times \frac{1}{(37)^{1/4}}\left(\frac{\mu}{\rho U x}\right)^{1/5}$$

$$= 0.0577 \times \frac{\rho U^2}{2}\left(\frac{\mu}{\rho U x}\right)^{1/5} \quad \ldots(13.42)$$

(iii) Drag force (F_D) on one side of the plate is

$$F_D = \int_0^L \tau_0 \times b \times dx = \int_0^L 0.0577 \times \frac{\rho U^2}{2} \times \left(\frac{\mu}{\rho U}\right)^{1/5} \frac{1}{x^{1/5}} \times b \times dx$$

$$= 0.0577 \times \frac{\rho U^2}{2} \times \left(\frac{\mu}{\rho U}\right)^{1/5} \times b \int_0^L x^{-1/5}\, dx$$

$$= .0577 \times \frac{\rho U^2}{2} \times \left(\frac{\mu}{\rho U}\right)^{1/5} \times b \times \left[\frac{x^{4/5}}{4/5}\right]_0^L$$

$$= .0577 \times \frac{5}{4} \times \frac{\rho U^2}{2}\left(\frac{\mu}{\rho U}\right)^{1/5} \times b \times L^{4/5} = 0.072 \times \frac{\rho U^2}{2} \times \left(\frac{\mu}{\rho U}\right)^{1/5} \times b \times L^{4/5}$$

(iv) Drag co-efficient, C_D is given by

$$C_D = \frac{F_D}{\frac{1}{2}\rho A U^2}, \qquad \text{where } A = L \times b$$

$$= \frac{.072 \times \frac{\rho U^2}{2} \times \left(\frac{\mu}{\rho U}\right)^{1/5} \times b \times L^{4/5}}{\frac{\rho U^2}{2} \times b \times l}$$

$$= 0.072 \times \left(\frac{\mu}{\rho U}\right)^{1/5} \cdot \frac{1}{L^{1/5}} = 0.072\left(\frac{\mu}{\rho U L}\right)^{1/5}$$

$$= \frac{.072}{R_{e_L}^{1/5}} \quad \ldots(13.43) \quad \left\{\because R_{e_L} = \frac{\rho U L}{\mu}\right\}$$

2m